

KIRUTIGA SRIKANDA PRABANNA BALAN

Menlo Park, CA | krithika151997@gmail.com | kirutigabalan.github.io | 512-239-8133

SUMMARY

Process engineer with end-to-end cleanroom process flow ownership in photolithography, plasma RIE, and wet etch, plus hands-on DOE, SPC, and FMEA for yield and variability reduction. Strong in defect metrology (Dektak 6M profilometry, optical microscopy, SEM-EDS) failure analysis using FMEA and for process window characterization and root cause analysis.

EXPERIENCE

Engineering Consultant — Canival Medical Inc.

Sep 2025 – Present

- Reduced design risk 70% through FMEA under ISO 13485, identifying high-RPN failure modes above 300 and driving root cause analysis before production release.
- Designed leaflets in SolidWorks and created dimensional part drawings from 3D part files for stent and other components of the system
- Applied Design for Reliability and Design for Manufacturing principles to evaluate mechanical and electrochemical material performance under operational stress.
- Integrated traceability systems including DIOM and engineering database across cross-functional engineering workflows to support process history and risk tracking.

Electrochemistry Intern — Electra

Jun 2025 – Aug 2025

- Qualified 25+ deposition additives through structured DOE, identifying a standout additive that delivered 2x improvement in deposition uniformity, directional growth control and process stability.
- Developed quantitative defect metrics to track deposition morphology variability and process repeatability across experimental conditions.
- Performed JMP based statistical analysis on deposition morphology and penetration-depth distributions to identify key process parameters driving instability.
- Tracked and optimized KPIs including pH, conductivity, bath temperature, and additive concentration to improve deposition consistency and dendrite suppression.

Graduate Research Assistant — University of Texas at Austin, Aerospace Engineering

Jan 2022 – May 2025

- Owned end-to-end photolithography process flow from 3D design in SolidWorks for rapid prototyping a custom PDMS mold in a 3D printer, spin coating through mask alignment, exposure, and development in a Class 100 cleanroom
- Achieved repeatable and reliable results for copper extraction efficiency by 125% using magnetic-field-assisted electrochemical methods in the custom microfluidic device produced.
- 4x fabrication throughput improvement and reduced process variability through DOE applied across the photolithography flow, narrowing the process window.
- Optimized plasma bonding on the March Plasma CS170IF RIE etcher, reducing exposure from 60 s to 22 ± 2 s and eliminating device leakage at the glass-PDMS interface.
- Performed defect inspection and step-height metrology using optical microscopy and the Dektak 6M stylus profilometer to characterize pattern fidelity and yield.

Graduate Research Assistant — University of Texas at Austin, Materials Science

Jan 2020 – Dec 2021

- Solved coupled PDE and ODE systems to identify key parameters driving degradation and failure mechanisms in a novel eutectic alloy anode for next generation batteries.
- 20% improvement in phase evolution prediction accuracy via cell-centered finite-difference modeling of eutectic alloy systems for materials process design.
- Developed GITT-based electrochemical characterization models to generate OCV profiles for anode lifetime prediction.

EDUCATION

M.S., Aerospace Engineering — University of Texas at Austin, 2022 – 2025

M.S., Materials Science and Engineering — University of Texas at Austin, 2020 – 2021

B.Tech., Mechanical Engineering — Amrita University, 2015 – 2019

- SAE SUPRA (2017): Designed steering subsystem in SolidWorks with GD&T-annotated drawings; authored part-level eBOM with materials, process, and fastener costing; conducted DFMEA across 6 subsystems reducing RPNs through design modifications and material analysis.

SKILLS

Process Engineering & Cleanroom: Photolithography (Suss MA6, Laurell), Plasma RIE (March Plasma CS170IF), HF Wet Etching, PDMS Soft Lithography, Class 100 Cleanroom, Process Window Characterization, Recipe Development

Metrology, Mechanical Design & Reliability: 3D Printing, Stylus Profilometry (Dektak 6M), Optical Microscopy, SEM-EDS, DOE, SPC (X-bar, I-MR), Root Cause Analysis, DFMEA, DfR, JMP, ISO 13485

Modeling & Software: COMSOL Multiphysics, MATLAB, Python (NumPy, SciPy, pandas, scikit-learn), PDE/ODE Systems